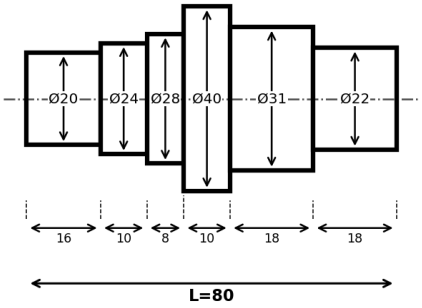
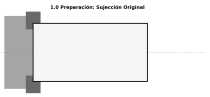
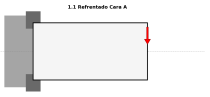
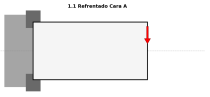
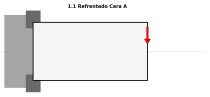
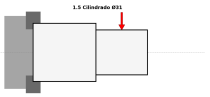
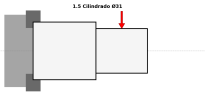
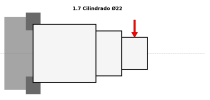
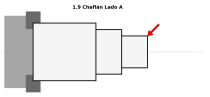
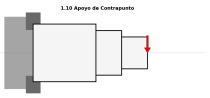
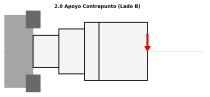
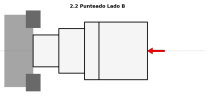
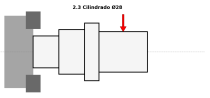
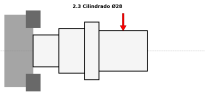
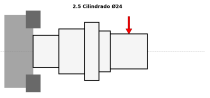
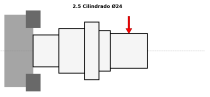
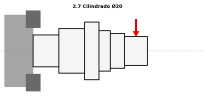
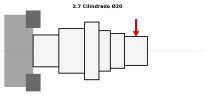
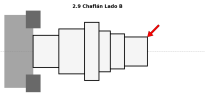
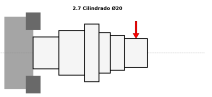
HOJA DE PROCESO 1ºME - TRINIDAD ARROYO	**CROQUIS DE LA PIEZA:**  	**P:** 01 **H:** 1/2
Alumno: Sergio Abad	**Material:** Acero AISI 1045	**Bruto:** Ø50 x 140 mm (Rev. A)

Nº Op.	Operación	Herramienta	Vc (m/min)	N (rpm)	a(mm/rev)	P (mm)	Croquis
1.0	Colocar tocho y alinear	—	—	—	—	—	
1.1	Refrentar cara frontal	Plaquette torneado derecha	120–160	800–1000	0.15–0.25	0.5–1	
1.2	**Punteado (centrado) para contrapunto**	Broca de centrar (Ø4–5 mm)	15–20	400–600	Manual / 0.05–0.10	3–5 mm total	
1.3	Cilindrar Ø40 (desbaste)	Plaquette desbaste	100–140	600–800	0.3–0.5	2–3 por pasada	
1.4	Cilindrar Ø40 (acabado)	Plaquette acabado	140–180	1000–1200	0.15–0.25	0.3–0.5	
1.5	Cilindrar Ø31 (desbaste)	Plaquette desbaste	100–130	700–900	0.3–0.5	2–3	
1.6	Cilindrar Ø31 (acabado)	Plaquette acabado	130–170	1000–1300	0.15–0.20	0.3	
1.7	Cilindrar Ø22 (desbaste)	Plaquette desbaste	90–120	800–1000	0.25–0.4	2	
1.8	Cilindrar Ø22 (acabado)	Plaquette acabado	120–160	1100–1400	0.12–0.20	0.3	

Nº Op.	Operación	Herramienta	Vc (m/min)	N (rpm)	a(mm/rev)	P (mm)	Croquis
1.9	Chaflanes / redondeos	Plaquette o herramienta chaflán 45°	100–140	800–1000	manual	—	
1.10	Colocar contrapunto (si no estaba)	Contrapunto fijo o giratorio	—	—	—	—	

Nº Op.	Operación	Herramienta	Vc (m/min)	N (rpm)	a(mm/rev)	P (mm)	Croquis
2.0	Colocar contrapunto en el punto existente	Contrapunto	—	—	—	—	
2.1	Refrentar cara final	Plaquette torneado	100–125	800–1000	0.15–0.25	0.5–1	
2.2	**Punteado (centrado) cara opuesta**	Broca de centrar (Ø4–5 mm)	15–20	400–600	Manual / 0.05–0.10	3–5 mm total	
2.3	Cilindrar Ø28 (desbaste)	Plaquette desbaste	90–120	700–900	0.3–0.5	2–3	
2.4	Cilindrar Ø28 (acabado)	Plaquette acabado	120–150	1000–1200	0.15–0.20	0.3	
2.5	Cilindrar Ø24 (desbaste)	Plaquette desbaste	80–110	800–1000	0.25–0.4	2	
2.6	Cilindrar Ø24 (acabado)	Plaquette acabado	110–140	1100–1300	0.12–0.20	0.3	
2.7	Cilindrar Ø20 (desbaste)	Plaquette desbaste	80–100	900–1100	0.25–0.35	1.5–2	
2.8	Cilindrar Ø20 (acabado)	Plaquette acabado	100–130	1200–1500	0.10–0.18	0.3	
2.9	Chaflanes / redondeos	Herramienta chaflán	80–110	800–1000	manual	—	

Nº Op.	Operación	Herramienta	Vc (m/min)	N (rpm)	a(mm/rev)	P (mm)	Croquis
2.10	Control dimensional final	Calibre, micrómetro, etc.	—	—	—	—